

Q.22 Write a note on battery and auxiliary power supply in electric trains.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain in detail the types of electric traction systems, with advantages and applications.

Q.24 Analyze the factors affecting schedule speed and acceleration in traction service. Provide examples.

Q.25 Discuss various control methods for DC traction motors and compare their efficiency.

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5th Sem/ Electrical Subject : Electrical Traction System

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 In an AC Locomotive, the primary. Source of power is derive from

- a) Ground rails b) Diesel Engine
- c) Batteries d) Overhead lines

Q.2 What is the purpose of the rheostatic control in DC series motors used in traction?

- a) Increase efficiency
- b) Improve braking
- c) Control starting current
- d) Control excitation

Q.3 Coasting is defined as :

- a) Running at full power
- b) Braking with full force
- c) Running without power supply
- d) Changing direction

- Q.4 Which braking method dissipates energy through resistors?
- Regenerative braking
 - Pneumatic braking
 - Rheostatic braking
 - Magnetic braking
- Q.5 What is the function of the tap changer in AC traction locomotives?
- Speed indication
 - Motor braking
 - Voltage control
 - Power factor correction
- Q.6 The conductor rail system is typically used in :
- High-speed trains
 - Metro and suburban systems
 - Goods trains
 - Diesel locomotives

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Write any two limitations of steam traction.
- Q.8 Define coasting in the context of traction.
- Q.9 Mention two desirable characteristics of traction motors.

- Q.10 What is chopper control?
- Q.11 Name one system where regenerative braking is widely used.
- Q.12 Write the function of lighting in a train coach.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Briefly describe the evolution of electric traction in India.
- Q.14 What are the different types of electric traction systems? Compare any two.
- Q.15 Draw a simplified speed-time curve and explain its key parts.
- Q.16 Explain how acceleration and braking affect schedule speed.
- Q.17 Compare DC shunt motor and DC series motor for traction usage.
- Q.18 What is rheostatic braking and where is it used?
- Q.19 Describe the function of thyristors in modern traction systems.
- Q.20 Explain emergency braking system and its application.
- Q.21 What are the key components of an OHE (Overhead Equipment) system?